

Assignment-2 Session8

Question 1: Pick any one feature from a popular app you use daily (for example: Instagram Signup, Flipkart Add-to-Cart, or Paytm Recharge) and write 5 clear requirements for that feature.

Answer

Application: Flipkart

Feature: Add to Cart

Requirement ID	Requirement Description
REQ-001	User should be able to add a product to the shopping cart by clicking the Add to Cart button.
REQ-002	Cart count should increase immediately after adding a product.
REQ-003	If the same product is added again, its quantity should increase instead of creating a duplicate item.
REQ-004	User should be able to remove a product from the cart successfully.
REQ-005	The cart should display the correct total price after every add or remove operation.

Question 2: For the requirements written in Task 1, create a table mapping each requirement to at least one test case.

Answer:

Requirement ID	Requirement Description	Test Case ID	Test Case Description
REQ-001	Add product to cart	TC-001	Verify that clicking Add to Cart adds the selected product to the cart.
REQ-002	Update cart count	TC-002	Verify that the cart count increases after adding a product.
REQ-003	Increase quantity	TC-003	Verify that adding the same product again increases its quantity.
REQ-004	Remove product	TC-004	Verify that clicking Remove deletes the product from the cart.
REQ-005	Update total price	TC-005	Verify that the total cart price updates correctly after adding or removing products.

Question 3: Extend your table by adding a 'Defect ID' column. Imagine that during testing, you find 2 bugs related to your test cases.

Answer

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Defect ID
REQ-001	Add product to cart	TC-001	Verify Add to Cart functionality	None
REQ-002	Update cart count	TC-002	Verify cart count updates	BUG-001
REQ-003	Increase quantity	TC-003	Verify quantity increases	None
REQ-004	Remove product	TC-004	Verify product removal	BUG-002
REQ-005	Update total price	TC-005	Verify total price calculation	None

Defect Details

BUG-001: Cart count does not increase after adding a product.

BUG-002: Clicking the **Remove** button does not remove the selected product from the shopping cart.

Question 4: Explain the difference between Forward Traceability and Backward Traceability. Give one example of each.

Answer

Forward Traceability

Forward Traceability ensures that every requirement has one or more corresponding test cases. It helps verify that all requirements are tested.

Example:

Requirement **REQ-002 (Cart count should increase)** is verified by **TC-002**.

Backward Traceability

Backward Traceability ensures that every test case is linked back to a valid requirement. It helps prevent unnecessary or extra test cases.

Example:

Test Case **TC-004 (Remove Product)** traces back to **REQ-004 (User should be able to remove a product from the cart)**.

Difference

Forward Traceability	Backward Traceability
Requirement → Test Case	Test Case → Requirement
Ensures all requirements are tested	Ensures every test case has a valid requirement
Prevents missing test coverage	Prevents unnecessary testing

Question 5: Create a Bi-Directional Requirements Traceability Matrix (RTM) for the Login feature of a food delivery application like Zomato.

Answer

Application: Zomato

Feature: Login Module

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Status
REQ-001	User should be able to log in using a valid email and password.	TC-001	Verify login with valid credentials.	Pass
REQ-001	User should be able to log in using a valid email and password.	TC-002	Verify login with an invalid password.	Pass
REQ-002	System should display validation messages for blank mandatory fields.	TC-003	Verify validation for blank email.	Pass
REQ-002	System should display validation messages for blank mandatory fields.	TC-004	Verify validation for blank password.	Pass
REQ-003	User should be able to reset the password using the Forgot Password option.	TC-005	Verify Forgot Password functionality.	Pass

Bi-Directional Traceability

Test Case ID	Linked Requirement
TC-001	REQ-001
TC-002	REQ-001
TC-003	REQ-002
TC-004	REQ-002
TC-005	REQ-003